Internship Report

COEP Technological University Pune

Data Science, Department of Computer Engineering

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1. Introduction

Autonomous and driver-assistance systems increasingly rely on understanding the fundamental reasons why drivers behave the way they do, moving beyond merely predicting what they will do next. While physical road testing is economically and practically infeasible for validating rare, safety-critical events, generative artificial intelligence models are increasingly used to synthesize virtual near-miss scenarios from naturalistic datasets, such as the NGSIM U.S. 101 highway trajectory data. However, standard generative models suffer from a significant vulnerability: they often learn spurious statistical correlations rather than true physical mechanisms. For example, when synthesizing a near-miss scenario, an unconstrained model might generate an ego vehicle's sudden deceleration based on irrelevant background variables rather than the actual kinematics of a decelerating lead vehicle, severely degrading the causal validity and safety value of the simulation.

Causal discovery offers a principled mathematical framework to recover these behavioral relationships directly from driving data, ensuring that models understand cause and effect rather than mere correlation. Yet, existing causal discovery approaches are heavily constrained by what a single vehicle's own sensors can observe. Due to occlusion and limited sensor range, critical variables that drive a vehicle's maneuvers often remain hidden in this ego-only regime, resulting in sparse and incomplete causal graphs. To overcome this limitation, Vehicle-to-Everything (V2X) cooperative perception provides a mechanism to extend a vehicle's awareness well beyond its immediate line of sight. By utilizing shared multi-agent kinematics — such as the preceding vehicle's instantaneous velocity, acceleration, and precise time headway — V2X-proxy features unlock a vastly richer contextual space. This project investigates whether this extended V2X context reveals critical causal structures in driving behavior that are completely invisible under isolated ego-only observation.

Recovering a multi-agent causal graph addresses only half of the challenge; ensuring that synthetic scenarios strictly adhere to these discovered physical rules requires architectural innovation. This research bridges the gap between causal inference and deep generative modeling by designing a mechanism capable of synthesizing near-miss driving scenarios constrained by a causal graph learned directly from empirical data. By translating the causal edges discovered via the constraint-based PC algorithm into a binary architectural mask applied to the decoder of a Conditional Variational Autoencoder (CVAE), the network is structurally forced to generate vehicle reactions attributable only to the correct, verified causal influences.

2. Objectives of the Internship

I. Causal Discovery Across Ego-Only and Cooperative-Perception Contexts

- **Core Goal:** To determine whether multi-agent, V2X-style context reveals causal relationships in driving behaviour that remain invisible when causal discovery is restricted to a single vehicle's own observations.
- **Methodological Approach:** This required constructing two parallel representations of the same naturalistic driving data: one limited strictly to a vehicle's own kinematics, and

one extended with the kinematics of neighbouring vehicles to act as a stand-in for cooperatively shared perception. Formal causal discovery was applied to both datasets so that the causal structure recoverable under each observation regime could be directly compared and quantified.

- **Research Target:** This objective specifically targets a limitation in existing causal-discovery research on driving behaviour: the field is largely confined to ego-only observation, even though modern cooperative-perception systems already make richer, multi-agent context available.

II. Causal-Graph-Constrained Generative Replay of Near-Miss Scenarios

- **Core Goal:** The second objective was to design a generative mechanism capable of synthesising near-miss driving scenarios whose internal agent-to-agent behaviour is mathematically constrained by a causal graph learned directly from data, rather than left for an unconstrained model to infer implicitly.
- **Methodological Approach:** This objective required building a conditional generative architecture whose internal connectivity mechanically reflects the previously discovered causal relationships. To establish a baseline, an architecturally identical unconstrained model was trained as a direct point of comparison. Both models were then evaluated on whether their generated reactions were attributable to the correct causal influences rather than to spurious correlation.
- **Research Target:** This addresses a critical gap identified in the surrounding literature, where near-miss scenario generators are routinely evaluated on realism and diversity, but rarely on whether the generated reactions are causally justified by the physical environment.

3. About Organization

The project was carried out within the Department of Data Science at COEP Technological University, Pune. COEP Technological University (formerly the College of Engineering, Pune) is one of India’s oldest and most established technical institutes, offering undergraduate and postgraduate programs across engineering and technology disciplines. The Department of Data Science supports applied and research-oriented coursework in machine learning, statistics, and data-driven systems, and provided the academic supervision and computing guidance under which this project was carried out. The department's coursework in causal inference, probabilistic modeling, and deep learning directly informed the methodology used in this project, and the project in turn served as an applied extension of that coursework to a real, open research problem.

4. Scope of Work

The scope of work encompassed the end-to-end design, implementation, and rigorous evaluation of a complex machine learning pipeline situated at the intersection of causal inference and generative artificial intelligence. Operating independently within the Department of Data Science at COEP Technological University, the responsibilities spanned from raw naturalistic data engineering to the architectural modification of deep generative models, all directed toward the broader social and technical initiative of enhancing autonomous vehicle safety validation.

The specific tasks, projects, and responsibilities undertaken included:

- Conducting an extensive literature review and synthesis of contemporary research across three converging threads: causal discovery in driving behavior, Vehicle-to-Everything (V2X) cooperative perception, and the generative replay of near-miss scenarios. This foundational task was critical to precisely identifying the technological gap that the project would address.
- Identifying and preparing the NGSIM U.S. 101 naturalistic driving dataset to support the project's dual causal-discovery and generative-modeling requirements. This involved adapting the data processing methodology to accommodate real-world constraints and managing the ingestion of highly granular trajectory data.
- Engineering multi-agent feature spaces by processing the raw trajectory data into two distinct representational views. This task required constructing an “Ego-only” feature set reflecting isolated vehicle sensors, and a “V2X-proxy” feature set that incorporated preceding-vehicle kinematics to simulate cooperatively shared perception.
- Formulating and extracting a highly specific subset of genuine near-miss driving events from the raw data. This required moving beyond simplistic definitions to implement a physically accurate, closing-speed-based true Time-To-Collision (TTC) criterion alongside a scaled hard-braking deceleration threshold.
- Applying formal causal discovery techniques, specifically executing the constraint-based PC algorithm independently on both the ego-only and V2X-proxy feature sets. This task culminated in a structured, quantitative comparison to determine the additional causal structure recovered when leveraging multi-agent context.
- Architecting and implementing a novel Causal-Graph-Constrained Conditional Variational Autoencoder (CVAE) using the PyTorch framework. A major technical responsibility was designing a custom masked linear layer that directly translated the discovered causal edges into a hardwired architectural constraint, forcing the generator to obey physical causal rules.
- Training both the newly designed causal-masked generator and an architecturally identical unconstrained baseline model to serve as a direct point of comparison.
- Evaluating the generative models through a rigorous quantitative framework designed specifically to isolate the effect of the causal constraint. This involved calculating and analyzing causal consistency scores, reconstruction accuracy, sample diversity, and single-scenario inference latency.

5. Activities Undertaken

The following comprehensive activities and technical milestones were undertaken:

- **Activity 1: Literature Synthesis and Research Gap Identification:** The project commenced with a rigorous literature review analyzing eight contemporary research works across three distinct domains: causal discovery in driving behavior, Vehicle-to-Everything (V2X) cooperative perception, and the generative synthesis of safety-critical scenarios. This activity culminated in the precise identification of a significant research gap: while unconstrained generative models are routinely evaluated for sample diversity and visual realism, they are rarely evaluated for causal faithfulness. This foundational phase directly shaped the project's dual innovation objectives.
- **Activity 2: Naturalistic Trajectory Data Acquisition and Pipeline Setup:** The Federal Highway Administration's NGSIM U.S. 101 highway trajectory dataset was acquired and justified as the foundational data source. The dataset, capturing complex multi-lane traffic behavior during a transitional traffic period (7:50 a.m. to 8:05 a.m.), was ingested into a custom data processing engine. A strict environment verification script was developed in Python to ensure all required machine learning and causal inference libraries (PyTorch, causal-learn, pandas) were correctly initialized.
- **Activity 3: Feature Engineering and V2X-Proxy Context Construction:** To directly compare observation regimes, a custom data preprocessing pipeline was engineered to process up to 300,000 raw trajectory frames. From this raw data, two parallel feature spaces were constructed. The first was an "Ego-only" feature set that extracted metrics such as mean speed, acceleration, and lane changes, representing what a single vehicle could observe through isolated sensors. The second was a "V2X-proxy" feature set, which artificially expanded the observation space by joining the preceding vehicle's kinematics (lead velocity, lead acceleration, and relative speed), successfully mimicking a connected-vehicle cooperative perception environment.
- **Activity 4: Near-Miss Mining and Mathematical Formulation:** Standard trajectory data contains mostly safe, mundane driving. To train a model on safety-critical events, a robust mathematical filter was designed to extract genuine near-miss events. An initial simple time-headway threshold was discarded in favor of a physically accurate, closing-speed-based true Time-To-Collision (TTC) metric. Furthermore, a hard-braking threshold of -10.0 ft/s^2 (approximately 0.3g) was established, ensuring that only genuine evasive maneuvers and imminent collision risks were passed to the generative modeling stage.
- **Activity 5: Causal Discovery and Graph Comparative Analysis:** The constraint-based PC algorithm ($\alpha = 0.05$) was applied independently to both the ego-only and V2X-proxy datasets. A key technical activity was structurally comparing the resulting directed acyclic graphs (DAGs) using network analysis libraries. This activity successfully proved the first research objective: the V2X-proxy graph recovered significantly more causal structure — expanding from 7 edges under ego-only observation to 18 edges

under V2X-proxy context, a gain of 12 causal links connecting context triggers to ego reactions that were completely missing in the ego-only graph. The resulting edge list was successfully exported to serve as the blueprint for the generative model.

- **Activity 6: Architecture Design of the Causal-Masked Generative Engine:** A major technical milestone was the design and implementation of a Conditional Variational Autoencoder (CVAE) from scratch using PyTorch. To enforce causal rules, a custom MaskedLinear network layer was coded. This layer ingested the discovered causal edge list and translated it into a binary weight mask, mechanically forcing the CVAE decoder to generate ego reactions exclusively from the context variables validated by the causal graph. For rigorous scientific comparison, a secondary, fully unconstrained baseline CVAE was also coded and trained under identical conditions.
- **Activity 7: Quantitative Evaluation and Performance Benchmarking:** The final activity involved designing an evaluation framework to quantitatively assess the generative models. A custom “Causal Consistency Gap” metric was formulated by calculating the absolute correlation matrices of the generated data and analyzing the delta between causally linked and unlinked variables. Alongside this, the models were evaluated for distributional reconstruction accuracy (Mean Absolute Error) and sample diversity to analyze the trade-offs of the causal constraint. Finally, single-scenario inference latency was benchmarked to verify the real-time deployment viability of the masked architecture.

6. Challenges Faced

Executing an end-to-end, multi-disciplinary machine learning pipeline independently naturally presents a spectrum of theoretical, technical, and logistical obstacles.

- **Algorithmic Translation and Architectural Mapping:** One of the most complex technical hurdles involved translating the causal graph — which was learned over aggregated, per-vehicle features — into a concrete, operational architectural constraint within the deep generative model. Because the causal-discovery phase and the generative-modeling phase operated at different levels of data granularity, direct integration was not natively supported by standard libraries. Mapping the discovered causal edges required engineering custom data structures to precisely align aggregated causal-graph node names (such as mean_accel) with the highly granular frame-level target features (such as v_Acc) required for CVAE training. Ensuring this causal mask matrix was perfectly aligned with the custom PyTorch MaskedLinear layer was critical; any misalignment in the tensor dimensions or feature indices would have caused the constrained model to either sever essential context pathways or degenerate entirely into an unconditional generator.
- **Physical Accuracy in Data Formulation and Unit Conventions:** A significant data engineering obstacle arose when attempting to correctly define and extract a genuine “near-miss” driving event from the raw trajectory logs. Initial detection criteria based on

simple time-headway proved vastly too permissive, artificially inflating the dataset with routine car-following frames rather than genuine safety-critical anomalies. Overcoming this required a mathematical redesign of the extraction logic to utilize a true, closing-speed-based Time-To-Collision (TTC) formulation, which only flagged events where the relative speed indicated an active collision course. Furthermore, processing the legacy NGSIM dataset demanded strict attention to unit-convention corrections. Because the original data was recorded in feet and feet-per-second, establishing a hard-braking threshold required precise scaling ($-10.0\ ft/s^2 \approx -3.05\ m/s^2$) to ensure the filtered events genuinely represented a 0.3g emergency evasive maneuver rather than routine highway deceleration noise.

- **Independent Project Execution and Hardware Optimization:** As a single researcher undertaking a full-stack data science project, independent project management, environment configuration, and debugging presented persistent workflow challenges. Operating without a team meant assuming simultaneous responsibility for data engineering, causal algorithmic tuning, deep learning architecture design, and rigorous statistical evaluation. Additionally, local resource constraints required explicit hardware optimization efforts. Training the generative models efficiently necessitated configuring the PyTorch environment to interface properly with an Apple Silicon M2 GPU via the MPS (Metal Performance Shaders) backend. Furthermore, when benchmarking the single-scenario inference latency for real-time deployment feasibility, the asynchronous nature of GPU execution distorted the initial timing metrics. This required programming manual hardware synchronization calls to force the GPU to finish execution before stopping the timer, ensuring the reported sub-millisecond benchmarking was scientifically valid and accurate.

7. Learning and Skill Development

The project served as a vital bridge between theoretical data science constructs and high-stakes, applied machine learning engineering. By executing an end-to-end research pipeline independently, the experience yielded profound advancements in technical proficiencies, soft skills, and strategic research methodologies, directly aligning with advanced academic coursework and future career aspirations in artificial intelligence.

Mastery of Applied Causal Inference and Deep Generative Modeling: A primary technical learning outcome was the successful transition from theoretical causal inference to applied structural learning. The project required deploying the constraint-based PC algorithm on highly noisy, real-world naturalistic trajectory data rather than pristine synthetic datasets. This built a deep intuition for how spatial-temporal variables interact and how multi-agent data resolves unobserved confounders in dynamic systems. Concurrently, the project fostered advanced expertise in deep generative modeling. Coding a Conditional Variational Autoencoder (CVAE) from scratch in PyTorch — and specifically engineering the custom MaskedLinear layer to enforce a discovered causal directed acyclic graph (DAG) — developed a highly specialized skill set in architecturally constrained neural networks. This proved that deep learning architectures can be explicitly programmed to obey physical and logical laws, rather than acting as uninterpretable black boxes.

Complex Data Engineering and Hardware-Optimized Execution: The project provided rigorous training in heavy-duty data engineering and feature extraction. Processing the NGSIM U.S. 101 trajectory dataset demanded writing efficient data pipelines capable of handling hundreds of thousands of frame-level kinematic records. Constructing the physically accurate, closing-speed-based true Time-To-Collision (TTC) metric solidified an understanding of how raw sensor logs must be mathematically transformed into features suitable for neural network ingestion.

Research Methodology and Quantitative Evaluation Design: Designing evaluation metrics capable of isolating the precise effect of an architectural modification developed strong scientific evaluation skills. Formulating the “Causal Consistency Gap” metric required writing custom statistical scripts to evaluate correlation matrices generated from synthetic outputs, teaching the importance of evaluating models not just for distributional accuracy (Mean Absolute Error), but for structural faithfulness. This capacity to mathematically prove a model's underlying behavioral logic is an essential skill for modern AI safety verification.

Independent Project Management and Career Alignment: From a soft-skills and professional development perspective, scoping an open-ended research problem to a fixed, defendable eight-week timeline independently cultivated strong project management, self-reliance, and debugging resilience. Operating without a team necessitated strict version control, reproducible environment management, and comprehensive documentation practices. Socially and professionally, the project instilled a deep appreciation for open-source safety methodologies; by building tools to synthesize causally accurate near-miss scenarios, the work actively contributes to the broader societal goal of deploying highly reliable and safe autonomous vehicles.

8. Impact Assessment

By bridging the gap between causal inference and deep generative modeling, the project produced tangible, measurable successes alongside broader societal contributions to autonomous vehicle safety.

Quantitative Indicators of Success and Tangible Outcomes: The project successfully established a highly measurable benchmark for causally faithful generative modeling. Through rigorous empirical evaluation, the research quantitatively demonstrated that V2X cooperative perception significantly enriches structural knowledge; the recovered causal graph expanded from a sparse 7 edges under ego-only observation to a dense 18 edges under the V2X-proxy regime — a gain of 12 newly recovered causal links — unlocking critical behavioral dependencies completely missed by isolated sensors. More importantly, the custom Causal-Graph-Constrained Conditional Variational Autoencoder (CVAE) achieved a Causal Consistency Gap of 0.4038, massively outperforming the unconstrained baseline model's score of 0.1297. This mathematical improvement proves that the architectural causal mask forces the network to synthesize near-miss scenarios with strict causal faithfulness rather than relying on spurious statistical correlations. Crucially, performance benchmarks recorded a single-scenario inference latency of just 0.1370 milliseconds using an Apple Silicon MPS hardware backend, definitively proving that this causally rigorous architecture is fully viable for real-time, hardware-in-the-loop autonomous vehicle simulation.

Qualitative Impact and Contributions to Social Causes: Beyond raw performance metrics, the project delivered substantial qualitative value to the Department of Data Science at COEP Technological University and the broader autonomous driving sector. Validating autonomous systems against rare, high-risk edge cases is a recognized global challenge, as capturing these events naturalistically is dangerously inefficient. This project provided a scalable, open-source software pipeline — released publicly under the MIT license — capable of artificially yet accurately generating these dangerous scenarios without risking physical human lives on public roads. By proving that causal graphs can be mechanically hardwired as architectural constraints, the research actively shifts the paradigm of AI safety verification from opaque, correlation-based synthesis to transparent, physically grounded scenario generation. This directly supports the societal and ethical imperative of ensuring driver-assistance and autonomous navigation systems are thoroughly and safely stress-tested before public deployment.

Impact on Personal Growth and Professional Development: On a personal and professional level, the project transformed theoretical academic concepts into robust, applied engineering expertise. Successfully managing the end-to-end development lifecycle — from processing the raw NGSIM U.S. 101 trajectory dataset to designing custom PyTorch neural network layers — fostered a profound degree of technical self-reliance. The necessity to independently scope the research gap, debug complex tensor dimension mappings, and optimize hardware execution environments instilled deep professional resilience and advanced problem-solving capabilities.

9. Conclusion

The project demonstrated that multi-agent, V2X-style context reveals causal structure in driving behaviour that is invisible to ego-only causal discovery, and that using this causal structure to constrain a near-miss scenario generator measurably improves the causal faithfulness of the generated scenarios relative to an unconstrained baseline. Both innovation objectives were met: a reusable method for recovering cooperative-perception causal structure from naturalistic data, and a causal-masked generative architecture that mechanically enforces the resulting causal graph during scenario synthesis. Taken together, the causal-discovery and generative-replay stages of the project form a connected pipeline in which discovered causal structure directly and measurably shapes the quality of generated safety-critical scenarios, rather than the two stages being evaluated separately, as is the case in the surrounding literature.

10. Recommendations

Based on the project experience, the following directions are recommended for further development:

- Extending the causal-masked generative approach from single-timestep reactions to full trajectory sequences, enabling generation of complete, temporally coherent near-miss manoeuvres rather than isolated reaction snapshots.
- Validating the V2X-context causal graphs developed in this project against genuine multi-sensor cooperative-perception data, to confirm that the causal relationships recovered here hold under real infrastructure and vehicle-to-vehicle sensor fusion.
- Applying the causal-masked generative architecture developed in this project to other safety-critical scenario-generation problems, such as pedestrian-interaction or intersection scenarios, where causally-justified agent behaviour is equally important.
- Scaling the causal-discovery stage to larger, multi-site naturalistic datasets, to test whether the discovered causal graphs remain consistent across different road geometries and traffic conditions.

11. The project was undertaken by a single student (team size: 1). The student was solely responsible for all aspects of the project, including literature review, dataset acquisition, implementation, evaluation, and reporting.

12. All data and software used in this project were obtained and used in accordance with their respective licenses and terms of use: the NGSIM dataset was used under its public terms of use for research and educational purposes, and all software dependencies (PyTorch, causal-learn, scikit-learn, and related libraries) are open-source. No personal, sensitive, or proprietary data was used or generated at any stage of the project. The project's codebase is released publicly under the MIT license, and all results reported reflect the project's actual measured outputs.

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